FEMA Disaster Declarations Dashboard — User Guide

🎯 Purpose

This Power BI dashboard provides interactive analysis of US FEMA disaster declarations across time, geography, incident types, and recovery programs. It supports decision-making for preparedness planning and resource allocation, and includes scenario simulation using What-If parameters.

🧭 Dashboard Navigation

The report is organized into the following pages:

Overview — Key KPIs including total disasters, average duration, and program activations
Temporal Trends — Yearly and monthly disaster trends
Seasonality — Monthly heatmap showing seasonal disaster patterns by state
Incident & Program Analysis — Disaster types vs assistance program usage
Geographic Impact — State-level disaster intensity map and top impacted states
Scenario Planning — What-If simulation of disaster frequency changes
Conclusion — Key insights and strategic recommendations

Use the page tabs at the bottom to navigate between sections.

🎛 Filters & Interactions

- Use slicers (Year, Incident Type) to filter visuals
- Clicking any chart element cross-filters other visuals
- Hover over visuals to view detailed tooltips
- Top-N visuals show highest-impact states or incident types for readability

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Seasonality Trends

1. **Hydrological & Flooding Events**

Water-related disasters caused by excess rainfall, river overflow, or dam issues.

- Flood
- Flash Flood
- Coastal Storm
- Dam/Levee Break
- Tsunami

📌 Rationale: Same impact pattern (water damage, displacement, infrastructure loss)

2. **Meteorological & Storm Events**

Atmospheric disturbances driven by weather systems.

- Hurricane
- Tropical Storm
- Severe Storm(s)
- Winter Storm
- Snowstorm
- Blizzard
- Ice Storm
- Hailstorm

📌 Rationale: All are storm-system driven and often declared together

3. **Geological Events**

- Earth-driven natural hazards.
- Earthquake
- Volcanic Eruption
- Landslide
- Mud/Landslide

📌 Rationale: Ground movement and seismic activity

4. **Fire Events**

Fire-related disasters, natural or weather-driven.

- Fire
- Wildfire

📌 Rationale: Same response strategy and damage type

5. **Climatological / Environmental Events**

Slow-onset or climate-driven disasters.

- Drought
- Heat Wave
- Freezing

📌 Rationale: Climate extremes rather than sudden shocks

6. **Human-Caused / Special Incidents**

Events not purely natural or with unique causes.

- Terrorist
- Other

Detailed Business Insights (Slide Wise)

Slide 1 – Overview

- This page presents the executive overview of FEMA disaster declarations.
- We summarize three key indicators: total disasters, average disaster duration, and total recovery program activations.
- The dataset shows approximately three thousand disaster declarations, indicating that disasters are recurring events requiring continuous preparedness.
- The average disaster duration is about twenty-four days, which tells us that disaster response operations typically extend across multiple weeks.
- We also observe that recovery programs are activated in a large share of disasters, indicating consistent federal assistance demand.
- Users can filter by year using the slicer to dynamically update all metrics.
- This overview gives decision makers a quick operational summary before moving into geographic and temporal deep-dive pages.

🔢 KPI 1 — Total Disasters = 3K

Meaning:

FEMA has issued approximately 3,000 disaster declarations in the dataset period.

Business Insight:

- Disaster declarations are **frequent**
- Indicates sustained national emergency activity
- Planning must be continuous — not seasonal only

Speak This:

This shows disasters are not rare events — they are recurring and require ongoing readiness planning.

⏱ KPI 2 — Average Duration = ~24 Days

Meaning:

On average, disasters remain active for about 3–4 weeks.

Business Insight:

- Resource deployment is **multi-week**
- Not short response — extended support needed
- Logistics + funding must be sustained

Speak This:

This indicates disaster response is not short-term — agencies must plan for multi-week operational support.

🏛 **KPI 3 — Program Activations = 3K**

Meaning:

Recovery programs (IA, PA, HM, IH) are triggered in most disaster events.

Business Insight:

- Most disasters require **federal assistance**
- Recovery funding demand is high
- Budget forecasting is critical

A high proportion of disasters trigger federal recovery programs, showing strong dependence on structured assistance mechanisms.

Slide 2 — TEMPORAL TRENDS REVIEW

This page tells: **How disaster declarations and program activations change over time**

Created DimDate - To enable proper time intelligence, correct month sorting, and star schema modeling for temporal analysis.

📈 **Insight — Long-Term Increase**

Disaster declarations increase significantly after the 1990s.

🔥 **Insight — Early 2000s Peak**

The early 2000s show the highest disaster and program activation spikes.

🔁 **Insight — Strong Correlation**

Program activations closely track disaster counts — showing responsive federal support.

📉 **Insight — Cyclical Pattern**

Activity rises in waves rather than steady growth — suggesting seasonal and cycle effects.

📊 **Business Insights From Monthly Chart**

Monthly trends show recurring peaks in specific periods, indicating seasonal disaster patterns such as hurricane and flood seasons.

Long-Term Trend

Disaster declarations increase significantly after the 1990s.

🔥 **Peak Activity**

The highest yearly disaster count reached about 187 declarations.

📅 **Monthly Pattern**

Monthly trends show wave-like patterns rather than steady distribution, indicating cyclical disaster behavior.

🏛 **Program Response Alignment**

Program activations closely follow disaster spikes, confirming responsive federal support mechanisms.

This page shows how disaster declarations evolve over time at both yearly and monthly levels. The top chart compares yearly disaster counts with program activations and shows a strong upward trend after the 1990s, with peak activity reaching around 187 disasters in a year. The monthly trend chart provides finer time resolution and shows cyclical wave patterns rather than uniform distribution. Together, these trends support seasonal readiness planning and long-term resource forecasting.

Slide 3 – Seasonality

🔥 **Peak Season Window**

Disaster declarations peak from June through September, with August highest.

❄️ **Low Season**

November through February show lowest disaster activity.

📈 **Risk Concentration**

Peak months show nearly 3× higher disaster frequency than winter months.

🧭 **Planning Insight**

Emergency staffing and funding should be ramped up before June.

📊 **Scatter Plot Insights**

Incident Group vs:

- Avg time to declaration
- Disaster volume

⏱️ **Fast Response — Low Delay Groups**

Climatological incidents show lower declaration delay.

🌪️ **High Volume Groups**

Meteorological & Hydrological incidents drive highest disaster counts.

🧠 **Analytical Insight Line**

Some incident groups are both high-frequency and slower to declare — indicating operational bottlenecks.

This page analyzes seasonal disaster patterns using a monthly heatmap by state. Darker cells indicate higher disaster activity. We observe a clear seasonal concentration between June and September, with August being the peak month overall. Winter months show significantly lower declaration counts. This confirms that disaster risk is seasonal and supports pre-season resource planning. The scatter plot compares incident groups by average declaration response time and disaster volume, showing that meteorological and hydrological incidents contribute the highest volumes.

Slide 4 - Incident and Program Analysis

BUSINESS INSIGHTS FROM YOUR ACTUAL VISUALS

🏛 **Program Demand Insight**

Public Assistance declarations (~37K) significantly exceed other programs, indicating infrastructure and community recovery dominates disaster support needs.

👤 **Household Impact**

Individual Assistance (~14K) shows a substantial share of disasters directly affect residents.

🏠 **Housing-Specific Support**

Individual Housing (~3K) is much lower, meaning only a subset of disasters require housing-focused intervention.

🌪 **Incident Drivers**

From bar + donut:

Floods, severe storms, and hurricanes account for the largest share of disaster declarations and program activations.

⏱ **Operational Complexity**

From scatter:

Some incident types show both longer durations and longer declaration delays, indicating higher operational complexity and slower response cycles.

This page analyzes disaster incidents and associated federal program declarations. The KPI cards summarize program activation volumes, where Public Assistance is the most frequently declared program, followed by Individual Assistance and Hazard Mitigation. The bar chart shows how different incident types drive different program declarations, with floods, severe storms, and hurricanes generating the highest program demand. The donut chart shows the distribution of disasters by incident type. The scatter plot compares incident types by average declaration response time and disaster duration, highlighting which incident categories are more complex and slower to resolve.

Slide 5 – Geographic impact

KPI Card — Most Impacted State

Insight: Texas has the highest number of FEMA disaster declarations across the dataset, indicating consistently high exposure to disaster events and relief program activation.

Business Meaning: High-risk state → priority for preparedness funding + mitigation planning.

🗺️ **Map — Disaster Declarations by State**

Insight: Disaster declarations are concentrated across central and southern US regions, with multiple high-frequency states forming a visible risk belt.

Business Meaning:

Regional clustering suggests:

- hazard corridors
- repeated vulnerability
- need for regional disaster readiness programs

📈 Top 10 States — Disaster Trends by Incident Group

Insight: Top states show different dominant incident groups — some driven by meteorological disasters, others by fire or hydrological events.

Business Meaning: Preparedness strategy must be **state-specific**, not national one-size-fits-all.

This page presents the geographic distribution of FEMA disaster declarations across the United States. At the top, we highlight the most impacted state based on total disaster declarations — which is Texas — indicating the highest exposure and program activation frequency. The filled map visual shows disaster intensity by state using color gradients, allowing quick identification of high-risk regions. Darker states represent higher disaster counts. When we hover on a state, we can also see supporting metrics such as average disaster duration and program declarations, giving operational context. On the right, we analyze the Top 10 states across incident groups over time. This shows that disaster drivers vary by state — some are dominated by meteorological events, while others show higher fire or hydrological incidents. This geographic view helps policymakers prioritize region-specific preparedness and targeted resource allocation.

Slide 6 – Scenario planning

What-If Slider

Insight: Allows simulation of disaster frequency change from −20% to +20%.

Business Meaning: Supports preparedness planning under changing climate or reporting conditions.

🟧 KPI — Total vs Projected Disaster Count

Insight: Base disaster count is ~3K. Under current selected adjustment, projected count shifts to ~2.91K.

Meaning: Even small % changes significantly affect operational load.

🟪 KPI — Projected Program Declarations

Insight: Program activations scale proportionally with disaster frequency.

Meaning: Funding and staffing demand grows non-linearly with event frequency.

📊 Adjusted Disaster Count by State

Insight: High-risk states like Texas remain top even after scenario adjustment.

Meaning: Relative risk ranking is stable — but volume changes.

📈 Program Activations vs Adjusted Totals

Insight: Program demand closely tracks disaster count — strong dependency.

Meaning: Scenario modeling helps budget forecasting.

This page demonstrates our What-If scenario modeling capability. We created a disaster frequency adjustment parameter that allows users to simulate increases or decreases in disaster occurrence. By moving the slider, all dependent measures — including projected disaster counts and program declarations — automatically recalculate using DAX measures. For example, at the current adjustment level, the projected disaster count changes from about 3,000 to approximately 2,910. We also see how projected program declarations scale with disaster frequency, which is important for funding and operational planning. The state-level adjusted chart shows that while total volumes change, high-risk states remain consistently high — indicating structural geographic risk. This scenario tool supports forward-looking resource planning and stress testing.

Slide 7 - Conclusion

- In conclusion, our analysis shows that disaster declarations are both temporally and geographically concentrated, with Texas emerging as the most impacted state.
- Public Assistance programs account for the largest share of declarations, and disaster duration averages about 24 days.
- Temporal and seasonal patterns indicate predictable peaks, which can support preparedness scheduling. Our scenario modeling demonstrates that even moderate increases in disaster frequency can significantly raise program demand.
- Therefore, we recommend targeted geographic preparedness, program-specific planning, and use of scenario forecasting for FEMA resource allocation.

🎚 What-If Scenario Parameter

The **Scenario Planning** page contains a disaster frequency adjustment slider.

How to use:

1. Move the slider to increase or decrease disaster frequency %
2. Projected disaster counts and program declarations update automatically
3. Use this to simulate resource demand under different risk scenarios

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🔄 Data Refresh

- To refresh data in Power BI Desktop:

Home → Refresh

- To refresh in Power BI Service:

Dataset → Refresh now

Dashboard visuals will update automatically after refresh.